PRIME MAGNITUDES AND NONVANISHING FOR A NONMULTIPLICATIVE INDEFINITE THETA SERIES OVER $\mathbb{Q}(\sqrt{5})$

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ABSTRACT. We study an independently defined norm-supported cone series $B(q) = \sum_{N\geq 0} B_N q^N$ over $\mathbb{Q}(\sqrt{5})$. Its coefficients are signed counts of points in two quadrants of an affine lattice coset, and the norm parametrization is $-\mathrm{N}(\beta(k,r)) = 10E(k,r)+1$. The sequence is not multiplicative: with $a(M) = B_{(M-1)/10}$ one has $a(11)a(31) = -2 \neq 3 = a(341)$. Nevertheless its values at rational primes are rigid. For every prime $p \equiv 1 \pmod{10}$ we construct a three-valued ideal invariant

$$\iota(\mathfrak{p}) = \delta(\pi) - \lambda(\pi) \in \mathbb{Z}/3\mathbb{Z},$$

where δ is an archimedean sector index and λ is the discrete logarithm of π in the group $(\mathcal{O}_K/2\mathcal{O}_K)^\times \simeq \mathbb{F}_4^\times$. The invariant combines an archimedean sector label with a congruence label; we do not identify it with a finite ray-class character. We prove a reflection identity, an exact one-atom lemma, and a sign-coherence lemma, yielding

$$B_{(p-1)/10} \in \{-2,-1,1,2\}, \qquad |B_{(p-1)/10}| = 2 \iff \iota(\mathfrak{p}) = 1.$$

In particular the prime coefficients never vanish. A specialization of the Hecke–Mitsui prime number theorem to twelve joint sector–congruence cells gives

$$\#\{p \leq X : p \equiv 1 \pmod{10},\ |B_{(p-1)/10}| = 2\} = \frac{1}{12}\,\mathrm{Li}(X) + o(X/\log X)$$

and the corresponding count for magnitude 1 is $\frac{1}{6}\,\mathrm{Li}(X) + o(X/\log X)$. Thus the relative densities are 1/3 and 2/3. The finite algebraic and combinatorial certificate consumers are machine-checked in Lean 4; the construction of the global certificates and the Hecke–Mitsui analytic input are not part of the formalization.

1. INTRODUCTION

The Andrews–Dyson–Hickerson and Cohen identities connect real-quadratic norm sums with Maass waveforms and multiplicative coefficients [1, 5, 6]. In several standard examples, the defining lattice or coset is stable under the fundamental totally positive unit, so the element sum descends to a Hecke-character sum over ideals and multiplicativity follows. This mechanism occurs in the original construction over $\mathbb{Q}(\sqrt{6})$ [1, 5], the $\mathbb{Q}(\sqrt{2})$ example of Corson–Favero–Liesinger–Zubairy [7], Lovejoy's overpartition series [11], the multiplicative q-hypergeometric families of Bringmann–Kane [2], and the double sums of Lovejoy–Osburn [12, 13]. The series considered here has the same broad norm-supported shape, but its defining coset is *not* unit-stable: it does not descend to a single Hecke-character sum over ideals, and its coefficients are not multiplicative. The resulting nontrivial unit phase is the structural feature studied below; no priority claim is needed for our arguments.

Let

$$K = \mathbb{Q}(\sqrt{5}), \qquad \mathcal{O}_K = \mathbb{Z}[\varphi], \qquad \varphi = \frac{1+\sqrt{5}}{2}, \qquad \varepsilon = \varphi^2 = \varphi + 1.$$

The unit φ has norm -1, while ε is the fundamental totally positive unit. We define an affine coset $L \subset \mathcal{O}_K$, a bijection $\beta : \mathbb{Z}^2 \to L$, and a cone-difference series B. The paper has three main goals:

(1) prove directly that the classical multiplicative mechanism is unavailable and give an explicit multiplicativity counterexample;
(2) classify the magnitudes of all coefficients at primes $p \equiv 1 \pmod{10}$ and prove nonvanishing;
(3) prove the absolute prime-counting asymptotics, and hence the relative densities $1/3$ and $2/3$, using joint sector-and-congruence equidistribution of prime elements.

The definition of the finite invariant combines an archimedean sector index with a congruence index. We do not prove that it factors through a finite ray-class quotient, and we do not use such a factorization. The density argument is instead formulated in terms of a joint sector-and-congruence prime-element asymptotic of Hecke–Mitsui type [8, 14, 10].

Here is the main result. The invariant ι is defined in Section 5.

Theorem 1.1 (Prime classification and density)**.** *For a rational prime $p \equiv 1 \pmod{10}$, put*

$$a(p) := B_{(p-1)/10}.$$

(These are the only primes that occur: the norm identity represents exactly the values $10N+1$, so $p = 5$ and the inert primes $p \equiv \pm 3 \pmod{10}$ never appear, and the primes $p \equiv 9 \pmod{10}$ split in $\mathbb{Q}(\sqrt{5})$ but are not of the form $10N+1$ and hence never enter the indexing.) Then:

(1) *$a(p) \in \{-2, -1, 1, 2\}$, so $a(p) \neq 0$;*
(2) *for either prime ideal $\mathfrak{p} \mid p$,*

$$|a(p)| = 2 \quad \Longleftrightarrow \quad \iota(\mathfrak{p}) = 1;$$

(3) *if $P_2(X)$ and $P_1(X)$ count the primes $p \le X$, $p \equiv 1 \pmod{10}$, for which $|a(p)|$ is respectively 2 and 1, then*

$$P_2(X) = \frac{1}{12}\,\mathrm{Li}(X) + o\left(\frac{X}{\log X}\right), \qquad P_1(X) = \frac{1}{6}\,\mathrm{Li}(X) + o\left(\frac{X}{\log X}\right).$$

Consequently the relative densities are $1/3$ and $2/3$.

The series was suggested by calculations around Chan's dissection of Θ_{10} [3, 4]. The present paper deliberately starts from the independent definition of B in Definition 2.7. A row-model factorization motivated that definition and has been checked through a finite coefficient range, but we do not claim here a source-faithful theorem identifying Chan's missing kernel with B; no result below depends on such an identification.

Section 2 gives the lattice and norm formulas. Section 3 studies the unit action and proves nonmultiplicativity. Sections 4 and 5 construct the sector–residue invariant. Section 6 proves the one-atom and sign-coherence lemmas and derives prime nonvanishing. Section 7 derives the density statement from its explicitly isolated analytic input. Section 8 records computational and formalization scope.

2. The lattice, cones, and norm parametrization

2.1. **Basic arithmetic of the quadratic field.** The conjugate of φ is $\bar{\varphi} = 1 - \varphi$. For $\alpha = a + b\varphi \in \mathcal{O}_K$,

$$\bar{\alpha} = (a+b) - b\varphi, \qquad \mathrm{Tr}(\alpha) = 2a + b, \qquad \mathrm{N}(\alpha) = a^2 + ab - b^2.$$

The two real embeddings are denoted by

$$\sigma_1(a + b\varphi) = a + b\varphi, \qquad \sigma_2(a + b\varphi) = a + b\bar{\varphi}.$$

Lemma 2.1 (Class number one)**.** *The ordinary and narrow class numbers of K are both one.*

Proof. The discriminant of K is 5. Minkowski's bound in degree two says that every ideal class contains an integral ideal of norm at most

$$\frac{2!}{2^2}\sqrt{5} = \frac{\sqrt{5}}{2} < 2.$$

The norm of a nonzero integral ideal is a positive integer, so the representative has norm 1 and is $\mathcal{O}_K$. Hence the ordinary class number is one. Since φ has norm -1, every principal ideal has a generator with either prescribed sign pattern, and the narrow class number is also one. □

Lemma 2.2 (Splitting of the relevant primes)**.** *Every rational prime $p \equiv 1 \pmod{10}$ splits as $p\mathcal{O}_K = \mathfrak{p}\bar{\mathfrak{p}}$ with two distinct principal prime ideals of norm p.*

Proof. Such a prime is odd and is not 5. Quadratic reciprocity gives

$$\left(\frac{5}{p}\right) = \left(\frac{p}{5}\right) = 1$$

because $p \equiv 1 \pmod 5$. Thus p splits in K. Principality follows from Lemma 2.1. □

2.2. **The affine coset and the exponent.**

Definition 2.3. Define

$$L := \{a + b\varphi \in \mathcal{O}_K : b - 3a \equiv 1 \pmod{10}\}.$$

This is a translate of the index-10 sublattice

$$L_0 := \{a + b\varphi : b - 3a \equiv 0 \pmod{10}\}.$$

Definition 2.4. For $k, r \in \mathbb{Z}$, define

$$\beta(k,r) := (r - 2k) + (4k + 3r + 1)\varphi$$

and

$$H(k,r) := 4k^2 + 2k + r^2 + (6k+1)r, \qquad E(k,r) := \frac{H(k,r)}{2}.$$

Define the two cones

$$\mathcal{A} := \{(k,r) \in \mathbb{Z}^2 : k \geq 0,\ r \geq 0\}, \qquad \mathcal{D} := \{(k,r) \in \mathbb{Z}^2 : k < 0,\ r < 0\}.$$

Lemma 2.5 (Integrality and cone nonnegativity)**.** *For every $k, r \in \mathbb{Z}$, the integer $H(k,r)$ is even, so $E(k,r) \in \mathbb{Z}$. Moreover $E(k,r) \geq 0$ on $\mathcal{A} \cup \mathcal{D}$.*

Proof. We have

$$H(k,r) = r(r+1) + 2k(2k+1) + 6kr,$$

a sum of two even products and an even multiple. Hence H is even. If $(k,r) \in \mathcal{A}$, then

$$E(k,r) = 2k^2 + k + 3kr + \frac{r(r+1)}{2} \geq 0.$$

If $(k,r) \in \mathcal{D}$, write $k = -m-1$ and $r = -s-1$ with $m, s \geq 0$. Then

$$E(-m-1, -s-1) = 2m^2 + 6m + 4 + 3ms + 3s + \frac{s(s+1)}{2} \geq 0.$$

No nonnegativity assertion is made outside the two cones. □

Proposition 2.6 (Bijection and norm identity)**.** *The map $\beta : \mathbb{Z}^2 \to L$ is a bijection. Its inverse is*

$$k = \frac{b - 3a - 1}{10}, \qquad r = a + 2k$$

for $a + b\varphi \in L$. Furthermore,

$$-\mathrm{N}(\beta(k,r)) = 10E(k,r) + 1.$$

Proof. For $\beta(k,r) = a + b\varphi$,

$$b - 3a - 1 = (4k + 3r + 1) - 3(r - 2k) - 1 = 10k,$$

so $\beta(k,r) \in L$. The displayed inverse formulas are integral for an element of L and recover k, r, proving bijectivity. Substituting $a = r - 2k$ and $b = 4k + 3r + 1$ into $\mathrm{N}(a + b\varphi) = a^2 + ab - b^2$ gives

$$\mathrm{N}(\beta(k,r)) = -10E(k,r) - 1.$$

□

Definition 2.7 (The cone series)**.** For $N \geq 0$, define

$$B_N := \sum_{\substack{(k,r)\in\mathcal{D} \\ E(k,r)=N}} (-1)^r - \sum_{\substack{(k,r)\in\mathcal{A} \\ E(k,r)=N}} (-1)^r, \qquad B(q) := \sum_{N\geq 0} B_N q^N.$$

For later use, define the contribution of a single atom by

$$w(k,r) := \begin{cases} -(-1)^r, & (k,r) \in \mathcal{A}, \\ (+1)(-1)^r, & (k,r) \in \mathcal{D}, \\ 0, & \text{otherwise.} \end{cases}$$

Lemma 2.8 (Finiteness of coefficients)**.** *For every $N \geq 0$, both sums in Definition 2.7 are finite.*

Proof. On $\mathcal{A}$ the expression in Lemma 2.5 gives $2k^2 + k \leq N$ and $r(r+1)/2 \leq N$. Hence k and r are bounded. On $\mathcal{D}$, the formula with $k = -m-1$, $r = -s-1$ gives $2m^2 \leq N$ and $s(s+1)/2 \leq N$, so m and s are bounded. □

Corollary 2.9 (Norm support)**.** *If $B_N \neq 0$, then $10N + 1$ is the norm of an algebraic integer of K.*

Proof. A nonzero coefficient contains an atom $\beta(k,r)$ of norm $-(10N+1)$. Multiplication by φ, whose norm is -1, changes this to norm $10N+1$. ☐

Remark 2.10 (Provenance). The row-model factorization that motivated B was verified coefficient-by-coefficient through $N = 2048$. This finite check is not a theorem identifying the original missing kernel with B, and the proof of the present paper does not use it.

3. Unit action and nonmultiplicativity

The congruence defining L has one component modulo 2 and one component modulo the ramified prime $\sqrt{5}$.

Lemma 3.1 (Two congruence descriptions of L)**.** *For $\alpha = a + b\varphi \in \mathcal{O}_K$, the following are equivalent:*

(1) $\alpha \in L$;
(2)

$$\alpha \equiv 3 \pmod{\sqrt{5}\mathcal{O}_K} \quad \textit{and} \quad \alpha \bmod 2 \in \{1, \varphi\} \subset \mathbb{F}_4^{\times}.$$

Proof. Modulo $\sqrt{5}$ one has $\varphi \equiv 3 \pmod 5$, so

$$\alpha \equiv a + 3b \pmod{\sqrt{5}}.$$

If $c = b - 3a$, then $a + 3b \equiv 3c \pmod 5$. Thus $c \equiv 1 \pmod 5$ if and only if $\alpha \equiv 3 \pmod{\sqrt{5}}$. Modulo 2, the condition $b - 3a \equiv 1$ is $a + b \equiv 1$, which says that exactly one of a, b is odd. These are precisely the residues 1 and φ in $\mathcal{O}_K/2\mathcal{O}_K \simeq \mathbb{F}_4$. Combining the mod-2 and mod-5 conditions by the Chinese remainder theorem gives the mod-10 condition. ☐

Proposition 3.2 (Unit instability and stabilizer)**.** *One has*

$$\varepsilon L \cap L = \varnothing.$$

Moreover, for $j \in \mathbb{Z}$,

$$\varepsilon^j L = L \quad \Longleftrightarrow \quad 6 \mid j.$$

Proof. Modulo $\sqrt{5}$, one has

$$\varepsilon = \varphi^2 \equiv 3^2 \equiv -1 \pmod 5.$$

Thus ε has order 2 in the $\sqrt{5}$ component. Since every element of L is congruent to 3 modulo $\sqrt{5}$, multiplication by ε changes that residue to $-3 \equiv 2$, proving $\varepsilon L \cap L = \varnothing$. More generally, preservation of the $\sqrt{5}$ condition requires j even.

Modulo 2, the group $\mathbb{F}_4^{\times}$ is cyclic of order 3 and $\varepsilon \equiv \varphi + 1$ is a generator. The two residues allowed by Lemma 3.1 have discrete logarithms 0 and 2. Translation of the set $\{0, 2\}$ by j preserves it if and only if $j \equiv 0 \pmod 3$. Hence $\varepsilon^j L = L$ exactly when j is divisible by both 2 and 3, equivalently by 6. ☐

The proposition explains why the standard one-coset descent to ideals is unavailable. It does not by itself prove that the coefficient sequence is nonmultiplicative; the following finite witness does.

Theorem 3.3 (Nonmultiplicativity). *For $M \equiv 1 \pmod{10}$ define*

$$a(M) := B_{(M-1)/10}.$$

Then

$$a(11) = 1, \qquad a(31) = -2, \qquad a(341) = 3,$$

and therefore

$$a(11)a(31) = -2 \neq 3 = a(11 \cdot 31).$$

Proof. The complete atom lists are

N	cone	(k, r)	$(-1)^r$	contribution to B_N
1	$\mathcal{A}$	$(0, 1)$	-1	$+1$
3	$\mathcal{A}$	$(0, 2)$	$+1$	-1
	$\mathcal{A}$	$(1, 0)$	$+1$	-1
34	$\mathcal{A}$	$(2, 3)$	-1	$+1$
	$\mathcal{D}$	$(-1, -6)$	$+1$	$+1$
	$\mathcal{D}$	$(-3, -2)$	$+1$	$+1$

For the A-cone, the inequalities in the proof of Lemma 2.8 bound k and r before substitution. For the D-cone, use $k = -m-1$, $r = -s-1$ and the corresponding bounds. Checking those finite ranges gives exactly the displayed solutions. Summing the contributions gives the three values and the claimed inequality. □

4. A half-open sector and the cone criterion

For a nonzero $\alpha \in K$ of negative norm, define

$$R(\alpha) := \left|\frac{\sigma_1(\alpha)}{\sigma_2(\alpha)}\right|, \qquad \rho := \varphi^4.$$

Both embeddings of ε are positive and

$$R(\varepsilon\alpha) = \rho R(\alpha).$$

We use the half-open fundamental sector

$$\mathcal{S} := \{\alpha : \sigma_1(\alpha) > 0 > \sigma_2(\alpha),\ 1 \leq R(\alpha) < \rho\}.$$

The opposite sector is $-\mathcal{S}$.

Lemma 4.1 (Prime elements do not lie on sector boundaries). *Let $p \neq 5$ be a rational prime and let $\alpha \in \mathcal{O}_K$ satisfy $\mathrm{N}(\alpha) = -p$. Then $R(\alpha) \neq \rho^j$ for every $j \in \mathbb{Z}$.*

Proof. If $R(\alpha) = \rho^j$, then $\gamma = \varepsilon^{-j}\alpha$ has $R(\gamma) = 1$. After replacing γ by $-\gamma$ if necessary, its embeddings have opposite signs and equal absolute value, so $\mathrm{Tr}(\gamma) = 0$. Writing $\gamma = a + b\varphi$, the equation $2a + b = 0$ implies that b is even and γ is an integral multiple of $\sqrt{5}$. Hence $\mathrm{N}(\gamma) = -5m^2$, which cannot equal $-p$ for $p \neq 5$. □

Definition 4.2 (Sector index)**.** Let $\alpha \in \mathcal{O}_K$ have norm $-p$ for a rational prime $p \neq 5$. By Lemma 4.1, there is a unique $\delta(\alpha) \in \mathbb{Z}$ such that

$$\rho^{\delta(\alpha)} < R(\alpha) < \rho^{\delta(\alpha)+1}.$$

Equivalently, $\varepsilon^{-\delta(\alpha)}\alpha$ has ratio strictly between 1 and ρ. We use the reduction of $\delta(\alpha)$ modulo 3 when forming ι.

The half-open convention fixes the boundary allocation in the general definition, and Lemma 4.1 shows that no relevant prime element is affected by it.

Lemma 4.3 (Behavior of the sector index)**.** *For every $m \in \mathbb{Z}$,*

$$\delta(\varepsilon^m\alpha) = \delta(\alpha) + m, \qquad \delta(-\alpha) = \delta(\alpha),$$

and

$$\delta(\bar{\alpha}) = -\delta(\alpha) - 1.$$

Proof. The first two identities follow from $R(\varepsilon^m\alpha) = \rho^m R(\alpha)$ and $R(-\alpha) = R(\alpha)$. For conjugation, $R(\bar{\alpha}) = R(\alpha)^{-1}$. If $\rho^d < R(\alpha) < \rho^{d+1}$, then

$$\rho^{-d-1} < R(\bar{\alpha}) < \rho^{-d},$$

which gives the last identity. □

The next proposition is the exact link between the geometric sector and the two cones in Definition 2.4.

Proposition 4.4 (Cone recognition)**.** *Let $\alpha = \beta(k, r)$. Then*

$$(k, r) \in \mathcal{A} \iff \sigma_1(\alpha) > 0 > \sigma_2(\alpha) \text{ and } 1 < R(\alpha) < \rho,$$
$$(k, r) \in \mathcal{D} \iff \sigma_1(\alpha) < 0 < \sigma_2(\alpha) \text{ and } 1 < R(\alpha) < \rho.$$

Proof. Write $\alpha = a + b\varphi$. Direct substitution gives

$$\mathrm{Tr}(\alpha) = 2a + b = 5r + 1$$

and, using $1 + \rho = 3\varphi^2$ and $\varphi + \rho\bar{\varphi} = -\varphi^2$,

$$\sigma_1(\alpha) + \rho\sigma_2(\alpha) = \varphi^2(3a - b) = -\varphi^2(10k + 1).$$

Suppose first that $\sigma_1(\alpha) > 0 > \sigma_2(\alpha)$. Then $R(\alpha) > 1$ is equivalent to $\mathrm{Tr}(\alpha) > 0$, hence to $5r + 1 > 0$, which for integral r is $r \geq 0$. Likewise $R(\alpha) < \rho$ is equivalent to $\sigma_1(\alpha) + \rho\sigma_2(\alpha) < 0$, hence to $10k + 1 > 0$, which is $k \geq 0$.

Conversely, if $k, r \geq 0$, then

$$\beta(k, r) = \varphi + 2k\sqrt{5} + r(1 + 3\varphi).$$

Each summand has positive first embedding and negative second embedding. The two displayed linear identities then give $1 < R(\alpha) < \rho$.

For the D-cone apply the same argument to $-\alpha$. The inequalities become $-(5r+1) > 0$ and $10k + 1 < 0$, equivalent to $r \leq -1$ and $k \leq -1$. Conversely, writing $k = -m - 1$, $r = -s - 1$ gives

$$-\beta(k, r) = (-1 + 6\varphi) + 2m\sqrt{5} + s(1 + 3\varphi),$$

whose summands again have positive first embedding and negative second embedding. This proves the second equivalence. □

5. The sector–residue invariant

Because 2 is inert in K,

$$\mathcal{O}_K/2\mathcal{O}_K \simeq \mathbb{F}_4, \qquad \mathbb{F}_4^\times = \langle \varepsilon \bmod 2\rangle.$$

For $\alpha \not\equiv 0 \pmod 2$, define

$$\lambda(\alpha) \in \mathbb{Z}/3\mathbb{Z}$$

by

$$\alpha \equiv \varepsilon^{\lambda(\alpha)} \pmod 2.$$

Thus

$$\lambda(\varepsilon^m\alpha) = \lambda(\alpha) + m, \qquad \lambda(-\alpha) = \lambda(\alpha).$$

Conjugation on $\mathbb{F}_4$ is Frobenius, so

$$\lambda(\bar{\alpha}) = 2\lambda(\alpha).$$

Definition 5.1 (The invariant ι)**.** Let $p \equiv 1 \pmod{10}$ be prime and let $\mathfrak{p} \mid p$. Choose a generator π of $\mathfrak{p}$ with $\mathrm{N}(\pi) = -p$, and define

$$\iota(\mathfrak{p}) := \delta(\pi) - \lambda(\pi) \in \mathbb{Z}/3\mathbb{Z}.$$

Proposition 5.2 (Well-definedness)**.** *The value $\iota(\mathfrak{p})$ is independent of the chosen generator of norm $-p$.*

Proof. The units of norm 1 are $\{\pm\varepsilon^m : m \in \mathbb{Z}\}$. Thus another generator of norm $-p$ has the form $\pi' = \pm\varepsilon^m\pi$. By Lemma 4.3,

$$\delta(\pi') = \delta(\pi) + m,$$

while the discrete logarithm satisfies

$$\lambda(\pi') = \lambda(\pi) + m \pmod 3.$$

Their difference is unchanged. □

Theorem 5.3 (Reflection identity)**.** *For conjugate prime ideals,*

$$\iota(\bar{\mathfrak{p}}) \equiv -\iota(\mathfrak{p}) - 1 \pmod 3.$$

Proof. Use $\bar{\pi}$ as a generator of $\bar{\mathfrak{p}}$. By Lemma 4.3, $\delta(\bar{\pi}) = -\delta(\pi) - 1$. In $\mathbb{F}_4$, $\lambda(\bar{\pi}) = 2\lambda(\pi)$. Hence

$$\begin{aligned}\iota(\bar{\mathfrak{p}}) &= (-\delta(\pi) - 1) - 2\lambda(\pi)\\ &\equiv -(\delta(\pi) - \lambda(\pi)) - 1 \pmod 3.\end{aligned}$$

□

Remark 5.4. The invariant has finite codomain, but its definition uses the archimedean sector label δ as well as a finite congruence label. We neither identify it with a ray-class character nor use a Chebotarev argument. Section 7 instead isolates the joint sector-and-congruence equidistribution statement needed below.

6. Prime atoms, uniqueness, and sign coherence

We first translate membership in L into the normalized sector data.

Lemma 6.1 (Normalization into L)**.** *Let $p \equiv 1 \pmod{10}$ be prime, let $\mathfrak{p} \mid p$, and let π generate $\mathfrak{p}$ with $\mathrm{N}(\pi) = -p$. Put*

$$\pi_0 := \varepsilon^{-\delta(\pi)}\pi.$$

Exactly one of π_0 and $-\pi_0$ is congruent to 3 modulo $\sqrt{5}$. Denote that choice by α. Then

$$\alpha \in L \iff \iota(\mathfrak{p}) \in \{0, 1\}.$$

When this holds, $\alpha = \beta(k, r)$ for a point in $\mathcal{A} \cup \mathcal{D}$.

Proof. Modulo $\sqrt{5}$, conjugation is trivial, so $\mathrm{N}(\pi_0) \equiv \pi_0^2$. Since $\mathrm{N}(\pi_0) = -p \equiv -1 \equiv 4 \pmod 5$, the residue of π_0 is 2 or 3. Exactly one sign therefore gives residue 3.

The sector normalization gives $1 < R(\alpha) < \rho$ by Lemma 4.1. Its discrete logarithm is

$$\lambda(\alpha) = \lambda(\pi) - \delta(\pi) = -\iota(\mathfrak{p}) \pmod 3.$$

By Lemma 3.1, membership in L is equivalent to the already imposed residue 3 modulo $\sqrt{5}$ together with $\lambda(\alpha) \in \{0, 2\}$. The latter condition is equivalent to $\iota(\mathfrak{p}) \in \{0, 1\}$. If $\alpha \in L$, write $\alpha = \beta(k, r)$ by Proposition 2.6; then Proposition 4.4 places (k, r) in the A- or D-cone according to the sign pattern of the embeddings. □

Theorem 6.2 (The unique bad class)**.** *A prime ideal $\mathfrak{p} \mid p$ contributes an atom to $B_{(p-1)/10}$ if and only if*

$$\iota(\mathfrak{p}) \neq 2.$$

Proof. This is exactly Lemma 6.1: the contributing classes are 0 and 1, while class 2 fails the mod-2 condition defining L. □

The next result supplies the multiplicity statement needed in the prime theorem.

Lemma 6.3 (One atom per contributing prime ideal)**.** *Let $p \equiv 1 \pmod{10}$ be prime and let $\mathfrak{p} \mid p$. Among all pairs $(k, r) \in \mathcal{A} \cup \mathcal{D}$ for which*

$$(\beta(k, r)) = \mathfrak{p}, \qquad \mathrm{N}(\beta(k, r)) = -p,$$

there is at most one. If $\iota(\mathfrak{p}) \neq 2$, there is exactly one.

Proof. Existence for $\iota(\mathfrak{p}) \neq 2$ is Lemma 6.1. For uniqueness, suppose $\alpha = \beta(k, r)$ and $\alpha' = \beta(k', r')$ are two such atoms. They generate the same ideal and have the same norm, so

$$\alpha' = \pm\varepsilon^m \alpha$$

for some $m \in \mathbb{Z}$. Proposition 4.4 gives

$$1 < R(\alpha) < \rho, \qquad 1 < R(\alpha') < \rho.$$

But $R(\alpha') = \rho^m R(\alpha)$, and two numbers in the same interval of multiplicative width ρ can satisfy this only when $m = 0$. Therefore $\alpha' = \pm\alpha$.

Both elements lie in L, hence both are congruent to 3 modulo $\sqrt{5}$ by Lemma 3.1. Negation changes that residue to $-3 \equiv 2$, so $\alpha' = -\alpha$ is impossible. Thus $\alpha' = \alpha$, and bijectivity of β gives $(k', r') = (k, r)$. □

When both conjugate prime ideals contribute, their two atoms cannot cancel. The proof gives an explicit coordinate relation.

Lemma 6.4 (Sign coherence)**.** *Let $p \equiv 1 \pmod{10}$ be prime. Suppose both $\mathfrak{p}$ and $\bar{\mathfrak{p}}$ contribute atoms to $B_{(p-1)/10}$. If $\alpha = \beta(k, r)$ is the atom belonging to $\mathfrak{p}$, then the atom belonging to $\bar{\mathfrak{p}}$ is*

$$\alpha' = -\varepsilon\bar{\alpha} = \beta\left(\frac{r}{2}, 2k\right).$$

In particular r is even, the two atom coordinates lie in the same cone, and their contributions to $B_{(p-1)/10}$ have the same sign.

Proof. If both ideals contribute, Theorem 6.2 and the reflection identity leave only the pair of classes

$$\iota(\mathfrak{p}) = \iota(\bar{\mathfrak{p}}) = 1.$$

For the normalized atom α, the sector index is 0, so

$$\lambda(\alpha) = -\iota(\mathfrak{p}) = 2 \pmod{3}.$$

Thus $\alpha \equiv \varepsilon^2 \equiv \varphi \pmod{2}$. If $\alpha = a + b\varphi$, this says that a is even and b is odd. Since $a = r - 2k$, the integer r is even.

Set $\alpha' = -\varepsilon\bar{\alpha}$. It generates $\bar{\mathfrak{p}}$ and has norm $-p$. Modulo $\sqrt{5}$, conjugation fixes the residue, $\varepsilon \equiv -1$, and $\alpha \equiv 3$, hence

$$\alpha' \equiv -(-1) \cdot 3 \equiv 3 \pmod{\sqrt{5}}.$$

Modulo 2, conjugation is Frobenius. Since $\alpha \equiv \varepsilon^2$, one has $\bar{\alpha} \equiv (\varepsilon^2)^2 = \varepsilon$, and therefore $\alpha' \equiv \varepsilon^2 \pmod{2}$. Lemma 3.1 gives $\alpha' \in L$.

Furthermore,

$$R(\alpha') = R(\varepsilon\bar{\alpha}) = \frac{\rho}{R(\alpha)},$$

which lies strictly between 1 and ρ. Hence α' is an atom by Proposition 4.4; uniqueness follows from Lemma 6.3.

It remains to compute its coordinates. Since

$$\bar{\alpha} = (a + b) - b\varphi,$$

multiplication by ε and negation give

$$-\varepsilon\bar{\alpha} = -a + (b - a)\varphi.$$

Applying the inverse formulas of Proposition 2.6,

$$k' = \frac{(b - a) - 3(-a) - 1}{10} = \frac{2a + b - 1}{10} = \frac{r}{2}, \qquad r' = -a + 2k' = 2k.$$

If (k, r) is in the A-cone, then $(r/2, 2k)$ is in the A-cone; if it is in the D-cone, then $(r/2, 2k)$ is in the D-cone. Both r and $r' = 2k$ are even. Thus both internal signs $(-1)^r$ equal $+1$, and because the cone is the same, Definition 2.7 assigns the same outer sign to both atoms. $\square$

Theorem 6.5 (Prime magnitudes and nonvanishing). *For every prime $p \equiv 1 \pmod{10}$ and either $\mathfrak{p} \mid p$,*

$$B_{(p-1)/10} \in \{-2,-1,1,2\},$$

and

$$\left|B_{(p-1)/10}\right| = 2 \iff \iota(\mathfrak{p}) = 1.$$

In particular $B_{(p-1)/10} \neq 0$.

Proof. Let $i = \iota(\mathfrak{p})$. By Theorem 5.3, the ordered pair of classes of the two conjugate ideals is

$$(0,2), \quad (1,1), \quad \text{or} \quad (2,0).$$

Theorem 6.2 says that class 2 contributes no atom and classes $0,1$ contribute. Lemma 6.3 says that each contributing ideal contributes exactly one atom. Thus in the first and third cases exactly one atom occurs, so the magnitude is 1. In the middle case two atoms occur, and Lemma 6.4 shows that they have the same sign, so the magnitude is 2. The middle case is exactly $i = 1$. In all cases at least one atom occurs and no cancellation to zero is possible. □

7. Density from Hecke–Mitsui equidistribution

We now prove the analytic part of Theorem 1.1. We keep prime ideals and prime elements separate, since the Hecke–Mitsui theorem naturally counts prime elements in an archimedean region with a simultaneous finite congruence condition.

Recall that

$$\mathcal{O}_K^\times = \{\pm\varphi^n : n \in \mathbb{Z}\}, \qquad \varepsilon = \varphi^2,$$

that the norm-one units are $\{\pm\varepsilon^m : m \in \mathbb{Z}\}$, and that the totally positive units are $\{\varepsilon^m : m \in \mathbb{Z}\}$. For every nonzero prime ideal $\mathfrak{p}$ not above 2 or 5, let $\pi_\mathfrak{p}$ be the unique generator satisfying

$$\mathrm{N}(\pi_\mathfrak{p}) = -\mathrm{N}(\mathfrak{p}), \qquad \sigma_1(\pi_\mathfrak{p}) > 0 > \sigma_2(\pi_\mathfrak{p}), \qquad 1 \leq R(\pi_\mathfrak{p}) < \rho.$$

Class number one first gives a generator. Multiplication by φ if necessary makes its norm negative, and multiplication by -1 fixes the sign pattern. The remaining associates preserving both choices are exactly $\varepsilon^m\pi_\mathfrak{p}$. Since $R(\varepsilon^m\alpha) = \rho^m R(\alpha)$, the half-open interval selects a unique one. By Lemma 4.1, a prime element on a boundary ray generates the prime above 5; away from that prime we may use the open inequalities $1 < R < \rho$.

Put

$$\mathfrak{m} = (2\sqrt{5}) = 2\sqrt{5}\,\mathcal{O}_K.$$

The ideals (2) and $(\sqrt{5})$ are coprime. Since 2 is inert and 5 is ramified, the Chinese remainder theorem gives

$$(\mathcal{O}_K/\mathfrak{m})^\times \simeq (\mathcal{O}_K/2\mathcal{O}_K)^\times \times (\mathcal{O}_K/\sqrt{5}\mathcal{O}_K)^\times \simeq \mathbb{F}_4^\times \times \mathbb{F}_5^\times, \qquad \#(\mathcal{O}_K/\mathfrak{m})^\times = 12.$$

For $\ell \in \mathbb{Z}/3\mathbb{Z}$ and $s \in \mathbb{F}_5^\times$, let $u_{\ell,s} \in (\mathcal{O}_K/\mathfrak{m})^\times$ be the unique class with

$$u_{\ell,s} \equiv \varepsilon^\ell \pmod{2\mathcal{O}_K}, \qquad u_{\ell,s} \equiv s \pmod{\sqrt{5}\mathcal{O}_K}.$$

Let $\widetilde{\Pi}_{\ell,s}(X)$ count prime ideals $\mathfrak{p} \nmid \mathfrak{m}$ of norm at most X whose canonical generator satisfies $\pi_\mathfrak{p} \equiv u_{\ell,s} \pmod{\mathfrak{m}}$.

Theorem 7.1 (Hecke–Mitsui input, specialized)**.** *For every $\ell \in \mathbb{Z}/3\mathbb{Z}$ and every $s \in \mathbb{F}_5^\times$,*

$$\widetilde{\Pi}_{\ell,s}(X) = \frac{1}{12}\,\mathrm{Li}(X) + o\left(\frac{X}{\log X}\right).$$

The same asymptotic holds after restricting to degree-one prime ideals.

Proof. Mitsui's generalized prime number theorem gives simultaneous archimedean and reduced-congruence equidistribution of prime elements; see [14], Corollary and Theorem on p. 35. For the precise modern form used here, see Kai's thin-cone theorem [10, Corollary 4.2.3]. Apply it with the fixed modulus $\mathfrak{m}$ and the open fundamental sector

$$\sigma_1(\pi) > 0 > \sigma_2(\pi), \qquad 1 < R(\pi) < \rho.$$

Partition its projectivized interval into finitely many sufficiently small open thin-cone charts and sum. The boundary contains no prime element away from the prime above 5, by the preceding normalization.

For each reduced class $u \in (\mathcal{O}_K/\mathfrak{m})^\times$, the resulting logarithmically weighted count has the form

$$\Theta_u(X) = CX + o(X),$$

where C is independent of u. Kai's refinement includes a possible exceptional-zero secondary term; because the modulus is fixed, it is $O(X^\beta)$ for a fixed $\beta < 1$ and hence is $o(X)$. The canonical-generator normalization identifies the union over all twelve reduced classes with the prime ideals away from $\mathfrak{m}$. Summing over those classes and applying the prime ideal theorem gives $12C = 1$. Thus $C = 1/12$, and partial summation gives the displayed asymptotic.

Finally, in a quadratic field every prime ideal of residue degree two and norm at most X lies over a rational prime at most $\sqrt{X}$. There are $O(\sqrt{X}/\log X)$ such ideals, so deleting them does not change the main term. $\square$

For a degree-one prime ideal $\mathfrak{p}$ of norm p, the canonical generator has $\mathrm{N}(\pi_\mathfrak{p}) = -p$. Conjugation is trivial modulo $\sqrt{5}$, so if $\pi_\mathfrak{p} \equiv s \pmod{\sqrt{5}}$, then

$$-p \equiv \mathrm{N}(\pi_\mathfrak{p}) \equiv \pi_\mathfrak{p}^2 \equiv s^2 \pmod{5}.$$

Consequently the degree-one prime ideals above rational primes $p \equiv 1 \pmod{10}$ are precisely the six selected cells

$$\ell \in \mathbb{Z}/3\mathbb{Z}, \qquad s \in \{2, 3\}.$$

The nonzero mod-2 component makes the norm odd, while $s = 2, 3$ is equivalent to $s^2 \equiv -1 \pmod{5}$. Conversely, both prime ideals above every $p \equiv 1 \pmod{10}$ have canonical generators in these six cells.

Proposition 7.2 (Equal distribution of ι on prime ideals)**.** *Let $I_i(X)$ count degree-one prime ideals of norm at most X above rational primes $p \equiv 1 \pmod{10}$ and satisfying $\iota = i \in \mathbb{Z}/3\mathbb{Z}$. Then*

$$I_i(X) = \frac{1}{6}\,\mathrm{Li}(X) + o\left(\frac{X}{\log X}\right) \qquad (i = 0, 1, 2).$$

In particular each value of ι has relative density $1/3$ among these prime ideals.

Proof. For the canonical generator $\pi_\mathfrak{p}$, the sector index is 0, so

$$\iota(\mathfrak{p}) = -\lambda(\pi_\mathfrak{p}) \pmod 3.$$

For each fixed ℓ there are exactly two selected cells, namely $s = 2$ and $s = 3$. Theorem 7.1 gives $\frac{1}{12}\operatorname{Li}(X)$ for each cell, hence $\frac{1}{6}\operatorname{Li}(X)$ for each value of $\iota = -\ell$. ☐

Theorem 7.3 (Rational-prime densities)**.** *Let $P_2(X)$ and $P_1(X)$ count rational primes $p \le X$ with $p \equiv 1 \pmod{10}$ for which respectively*

$$\left|B_{(p-1)/10}\right| = 2 \qquad \text{and} \qquad \left|B_{(p-1)/10}\right| = 1.$$

Then

$$P_2(X) = \frac{1}{12}\operatorname{Li}(X) + o\!\left(\frac{X}{\log X}\right),$$

and

$$P_1(X) = \frac{1}{6}\operatorname{Li}(X) + o\!\left(\frac{X}{\log X}\right).$$

Consequently, relative to rational primes $p \equiv 1 \pmod{10}$,

$$\operatorname{dens}\left(\left|B_{(p-1)/10}\right| = 2\right) = \frac{1}{3}, \qquad \operatorname{dens}\left(\left|B_{(p-1)/10}\right| = 1\right) = \frac{2}{3}.$$

Proof. Every such rational prime has exactly two conjugate prime ideals. The reflection map on classes is

$$i \longmapsto -i-1,$$

so it pairs 0 with 2 and fixes 1. By Theorem 6.5, magnitude 2 occurs exactly for the conjugate class pair $(1,1)$, whereas magnitude 1 occurs for $(0,2)$ or $(2,0)$. Consequently

$$I_1(X) = 2P_2(X) + O(1).$$

Proposition 7.2 gives

$$P_2(X) = \frac{1}{12}\operatorname{Li}(X) + o\!\left(\frac{X}{\log X}\right).$$

A magnitude-1 prime has one class-0 and one class-2 ideal, so

$$I_0(X) = I_2(X) = P_1(X) + O(1),$$

which gives the second asymptotic. Equivalently,

$$\#\{p \le X : p \equiv 1 \pmod{10}\} = \frac{1}{4}\operatorname{Li}(X) + o\!\left(\frac{X}{\log X}\right).$$

Dividing the two magnitude counts by this total gives the relative densities 1/3 and 2/3. ☐

Theorems 6.5 and 7.3 prove Theorem 1.1.

8. Computational checks and Lean scope

8.1. **Computational checks.** A direct termwise comparison of the two defining cones, signs, and exponents gives the exact identification, in Hickerson–Mortenson notation [9], $B = -f_{1,3,4}(X, -X^3, X)$. The coefficient implementation was also checked through $N = 200$. The following statements are computational checks, not additional inputs to the proofs above. The ranges and counts below are reproduced by the repository script `numerical_audit.py`.

- The motivating row-model factorization has been verified coefficient-by-coefficient through $N = 2048$. This computation is not used as a source-faithful identification with Chan's missing kernel.
- The prime classification has been checked for all 19,617 primes $p \equiv 1 \pmod{10}$ with $p \leq 1{,}000{,}001$ (equivalently $(p-1)/10 \leq 100{,}000$). No zero occurs in that exact range.
- The explicit values at 11, 31, and 341 are proved in Theorem 3.3; they are not being used merely as numerical evidence.
- The one-atom and sign-coherence lemmas were additionally verified by direct ideal-level enumeration for every prime $p \equiv 1 \pmod{10}$ with $p < 20{,}000$ (and sampled up to 10^6): each contributing prime ideal received exactly one cone atom, and paired contributions always carried equal signs. In particular the cases $p = 11, 31, 41, 61, 71, 101$ can be checked by hand.
- Nonmultiplicativity is systematic: among the coprime pairs of distinct primes $p < q$, $p, q \equiv 1 \pmod{10}$, with $pq \leq 10^6 + 1$ (7165 pairs), none satisfies $a(pq) = a(p)a(q)$.
- Through $N \leq 10^5$ one has $\max_N |B_N| = 7$, and among the indices in that range at which $10N + 1$ is a norm there are 4904 with $B_N = 0$, all of them composite values of $10N + 1$; empirically $|B_N| = O(d(10N + 1))$.

8.2. **Lean scope.** The accompanying Lean 4 development has no admitted lemmas in the finite algebraic and combinatorial layer it covers. It checks the integer-coordinate algebra for E and β, the norm identity, the coset congruences, the unit calculations ($\varepsilon L \cap L = \varnothing$, the ε^6 stabilizer), the finite $\mathbb{F}_4^\times$ arithmetic of λ including the Frobenius half of the reflection identity, the mod-3 obstruction underlying nonvanishing (relative to sector labels, which enter the development as certificate inputs rather than as proved geometric facts), the explicit nonmultiplicativity witness $B_1 = 1$, $B_3 = -2$, $B_{34} = 3$, the resulting theorem `BCoeff_not_multiplicative_witness`, and the completeness of three explicitly bounded atom enumerations. The coefficient B_{34} is checked by kernel evaluation; all of these results depend only on Lean's usual three axioms.

Machine-checked additionally, in the modules `Ch10_ConeRecognition`, `Ch10_OneAtom`, `Ch10_SignCoherence`, and `Ch10_PrimeBridge` (built on the project server with only the standard three axioms): the integral coordinate-sign form of the cone-recognition identities, the one-atom lemma in associate form (two cone atoms that are associates by $\pm\varepsilon^m$ coincide), the conjugate-atom identity $-\varepsilon\bar{\alpha} = \beta(r/2, 2k)$ with its involution, cone-preservation, and evenness statements, and the bridge from atom counts to the coefficients — the classification $a(p) \in \{-2, -1, 1, 2\}$ and nonvanishing, relative to explicit per-prime certificate data.

More precisely, `SectorCert x` contains only a label $\delta \in \mathbb{Z}/3\mathbb{Z}$; it does not certify the proposition `InSector` δ `x`. A `SplitPrimeCert p` supplies a chosen generator π, primality and the congruence $p \equiv 1 \pmod{10}$, a norm equation $\mathrm{N}(\pi) = \pm p$, and the nonzero mod-2 condition. A `PrimeAtomCertificate` then supplies the common atom sign, the two finite sign-coherence fields, the two-atom bound, and the implications from a contributing iota class to an actual atom. Lean proves the coefficient classification and nonvanishing from these data; it does not construct these certificates uniformly for every rational prime.

Once these fields are supplied, the endpoint reduces to finite sign coherence, the bounds $1 \le$ `atomCount` ≤ 2, and the coefficient bridge. The primality and sector labels do not by themselves construct the certificate or add a global existence theorem. Thus the Lean result is a conditional finite bookkeeping theorem, not a proof of prime nonvanishing without hypotheses.

The following are *not* formalized: the geometric sector theory of Section 4 (including the geometric half of the reflection identity and the same-ideal-to-associate step, which uses class number one), the Hecke–Mitsui prime number theorem, the asymptotic density argument of Section 7, the uniform construction of `SplitPrimeCert` and `PrimeAtomCertificate` for every relevant prime, and a source-faithful bridge from Chan's missing kernel to B. The present paper proves the finite algebraic consequences of the supplied certificates and keeps the remaining geometric, global-existence, and analytic inputs explicit.

9. Concluding remarks

The series B shows that a norm-supported real-quadratic cone sum can retain rigid prime behavior after the standard multiplicative model has failed. The obstruction to the usual descent is visible in the unit action on the affine coset: εL is disjoint from L, and only ε^6 stabilizes the full coset. The explicit $11 \cdot 31$ calculation proves nonmultiplicativity, while the unit calculation explains why the classical one-coset construction is unavailable.

At primes, the replacement for the unavailable one-coset Hecke character model is the sector–residue invariant ι. Its reflection law gives the three possible conjugate class pairs. The one-atom lemma turns each good ideal into exactly one lattice point, and sign coherence prevents the pair in class $(1, 1)$ from cancelling. This yields nonvanishing and the two possible magnitudes. The specialized joint prime-element theorem gives the exact density of each of the twelve sector–congruence cells. The six cells over primes $p \equiv 1 \pmod{10}$ are therefore equidistributed, and explicit two-ideal bookkeeping transfers the resulting counts to rational primes.

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